

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

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COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour

Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

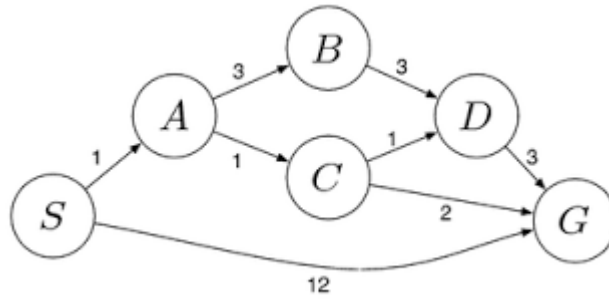
Signature of the student:

Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
 - i. What are the percept's for this agent?
 - ii. Characterize the operating environment.
 - iii. What are the actions the agent can take?
 - iv. How can one evaluate the performance of the agent?
 - v. What sort of agent architecture do you think is most suitable for this agent? Why?
3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.
4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.
5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse S to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

ANSWERS

1.

To Represent the amount of water in the **4-gallon jug** and **3-gallon jug** as **(4-gallon, 3-gallon)**.

Step	Action	State(4-gal,3-gal)
1	Fill the 3-gallon jug	(0, 3)
2	Pour the 3-gallon jug into the 4-gallon jug	(3, 0)
3	Fill the 3-gallon jug again	(3,3)
4	Pour water from the 3-gallon jug into the 4-gallon jug until the 4-gallon jug is full (only 1 gallon is needed)	(4,2)
5	Empty the 4-gallon jug onto the ground	(0,2)
6	Pour the remaining 2 gallons from the 3-gallon jug into the 4-gallon jug	(2,0)

Result: The **4-gallon jug** now contains exactly **2 gallons of water**, which is the required solution.

2.

i. Percepts are the information the Mars Rover receives from its sensors to understand its surroundings.

The Mars Rover uses:

- Cameras – capture images of rocks, soil, and the landscape.
- Temperature sensors – measure atmospheric and surface temperature.
- Weather sensors – detect wind speed, pressure, and humidity (where applicable).
- Chemical sensors/Spectrometers – identify the composition of rocks and soil.
- Navigation sensors (GPS is unavailable on Mars) – use IMU, wheel encoders, sun sensors, and visual odometry to estimate position.
- Battery and power sensors – monitor energy levels.
- Obstacle detection sensors – identify rocks, cliffs, and sand that could block **movement**.

ii. The Mars Rover operates in the following environment:

Environment Property	Description
Partially Observable	The rover cannot see or know everything around it at once.
Deterministic	Most actions have predictable results, though terrain conditions can introduce uncertainty.
Sequential	Each action affects future actions and mission success.
Dynamic	The environment changes due to dust storms, temperature changes, and shifting terrain.
Continuous	Movement, time, and sensor readings are continuous rather than discrete.
Single-Agent	The rover performs its mission independently without competing agents.
Unknown	The rover explores regions that have not been fully mapped beforehand.

iii. The Mars Rover can perform the following actions:

- Move forward, backward, left, or right.
- Navigate around rocks, craters, and other obstacles.
- Capture high-resolution photographs.
- Drill into rocks to collect samples.
- Analyze soil and rock samples using onboard scientific instruments.
- Measure weather and atmospheric conditions.
- Store collected samples.
- Transmit scientific data and images back to Earth.
- Recharge batteries using solar panels (for solar-powered rovers).
- Stop or enter a safe mode if dangerous conditions are detected.

iv. The rover's performance can be evaluated based on:

- Successfully reaching exploration targets.
 - Collecting and analyzing accurate rock and soil samples.
 - Avoiding obstacles and preventing damage.
 - Efficient use of battery power.
 - Sending reliable scientific data and images to Earth.
 - Completing mission objectives within the planned time.
 - Operating safely in harsh Martian conditions.
 - Maximizing scientific discoveries during the mission.
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3.

8-Queens Problem – Problem Formulation and Search Space

The 8-Queens Problem is a classic Artificial Intelligence search problem.

Problem Statement:

Place 8 queens on an 8×8 chessboard so that no two queens attack each other.

A queen can attack:

- Horizontally (same row)
- Vertically (same column)
- Diagonally

The goal is to arrange the queens so that none of them share the same row, column, or diagonal.

1. Problem Components

a) Initial State

- The chessboard is empty.
- No queens are placed.

Representation:

.....
.....
.....
.....
.....
.....
.....
.....

b) State Space

A state is any arrangement of queens on the board.

For efficient searching:

- Place one queen per column.
- Each new state adds one more queen.

Example:

Q

.. Q

.... Q . . .

. Q

This is a state with four queens.

c) Actions (Operators)

From any state, the possible actions are:

- Place a queen in the next column.
- Choose a row where the queen is not attacked.

Example:

If queens are placed in Columns 1, 2, and 3,

Next action:

Place Queen in Column 4

Choose any safe row.

d) Goal Test

The goal is reached when:

- All 8 queens are placed.
- No two queens attack each other.

That means:

- No same row
- No same column
- No same diagonal

e) Path Cost

Each queen placement costs 1.

Total cost

8

since eight queens are placed.

2. Search Space

The search space consists of all possible board configurations.

Without any restrictions:

$$8^8 = 16,777,216$$

possible arrangements.

Using the rule one queen per column, the search reduces to:

$$8! = 40,320$$

possible arrangements.

AI search algorithms further prune invalid states where queens attack each other, making the search much more efficient.

3. Search Strategy

Breadth-First Search (BFS)

- Explores all states level by level.
- Guarantees finding a solution.
- Requires a large amount of memory.

Depth-First Search (DFS)

- Places queens one by one.
- If a conflict occurs, it backtracks.
- Uses much less memory than BFS.
- Commonly used to solve the 8-Queens problem.

Backtracking Search

Steps:

1. Place the first queen in Column 1.
2. Move to Column 2.
3. Place the queen in a safe row.
4. Continue until all 8 queens are placed.
5. If no safe row exists, remove the previous queen (backtrack).
6. Try another row.
7. Repeat until a valid solution is found.

4. Example Solution

One valid arrangement is:

Q.....
....Q...
.....Q
.....Q..
..Q.....
.....Q.
.Q.....
...Q.....

Or as row positions by column:

Column	Row
1	1
2	7
3	5
4	8
5	2
6	4
7	6
8	3

4.

A Utility-Based Agent chooses the **best option** among many available choices by considering factors such as:

- Cost (fare)
- Comfort
- Travel time
- Cab availability
- Customer preference

For example, if the customer prefers **low cost**, the agent may suggest **Micro** or **Shared**. If the customer prefers **comfort**, it may recommend **Sedan** or **Prime**.

Thus, the agent selects the cab that provides the **highest utility**

Pseudocode:

Algorithm: OLA Cab Booking

START

Input Pickup_Location

Input Destination

Display Available Cab Types

1. Mini
2. Micro
3. Sedan
4. Shared
5. Prime

Input Customer_Choice

IF Customer_Choice is available THEN

 Calculate Fare
 Display Estimated Time
 Display Driver Details
 Confirm Booking
 Book the Cab
 Display "Cab Booked Successfully"

ELSE

 Display "Selected Cab Not Available"
 Suggest Another Cab Type

ENDIF

END

Working of the Utility-Based Agent:

- Customer enters the pickup and destination locations.
 - The agent checks the available cab types.
 - It compares fare, travel time, comfort, and availability.
 - It recommends the best cab according to the customer's preference.
 - After confirmation, the cab is booked.
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5.

From the given graph:

- $S \rightarrow A = 1$
- $S \rightarrow G = 12$
- $A \rightarrow B = 3$
- $A \rightarrow C = 1$
- $B \rightarrow D = 3$
- $C \rightarrow D = 1$
- $C \rightarrow G = 2$
- $D \rightarrow G = 3$

Cost Calculation

Path 1

$S \rightarrow G$

Cost = 12

Path 2

$S \rightarrow A \rightarrow B \rightarrow D \rightarrow G$

$= 1 + 3 + 3 + 3$

$= 10$

Path 3

$S \rightarrow A \rightarrow C \rightarrow D \rightarrow G$

$= 1 + 1 + 1 + 3$

$= 6$

Path 4

$S \rightarrow A \rightarrow C \rightarrow G$

$= 1 + 1 + 2$

$= 4$

Least-Cost Path: $S \rightarrow A \rightarrow C \rightarrow G$

Minimum Cost: 4

Uniform Cost Search always selects the path with the **smallest accumulated cost**. It ignores the number of steps and focuses only on the **total travel cost**. Among all possible routes, $S \rightarrow A \rightarrow C \rightarrow G$ has the minimum cost of 4, so UCS returns this as the optimal solution.
